

Prob 1 Induction:

$$(1) \sum_{k=0}^n (-1)^k \binom{n}{k} 2^{nk} = 1$$

$$\text{LHS} = \sum_{k=0}^n \binom{n}{k} (-1)^k \cdot 2^{nk} = (-1+2)^n = 1^n = 1$$

$$(2) \sum_{k=0}^n (-1)^k \binom{n}{k} 3^{nk} = 2^n$$

$$\text{LHS} = \sum_{k=0}^n \binom{n}{k} (-1)^k 3^{nk} = (-1+3)^n = 2^n$$

$$(3) \sum_{k=1}^{n+1} \frac{1}{k} \binom{n}{k-1} = \frac{2^{n+1}-1}{n+1}$$

$$\text{LHS} = \sum_{k=1}^{n+1} \frac{1}{k} \binom{n}{k-1}$$

$$\begin{aligned} &= \sum_{k=1}^{n+1} \frac{\binom{n}{k-1}}{n+1} = \frac{1}{n+1} (\binom{n}{0} + \dots + \binom{n}{n}) \\ &= \frac{1}{n+1} (2^{n+1} - 1) \quad \# \end{aligned}$$

$$\begin{aligned} k \binom{n}{k} &= n \binom{n-1}{k-1} \\ \frac{\binom{n}{k}}{n+1} &= \frac{\binom{n}{k}}{k} \end{aligned}$$

$$\text{P251 T23 (1)} \sum_{k=0}^m \binom{n-k}{m-k}$$

$$C_n^k = C_{n-1}^k + C_{n-1}^{k-1}$$

$$= \sum_{k=0}^m C_{n-k}^{m-k} = \sum_{k=0}^m C_{n-k}^{n-m} = C_n^{n-m} + C_{n-1}^{n-m} + \dots + C_{n-m}^{n-m}$$

$$= C_{n-m+1}^{n-m+1} + C_{n-m+1}^{n-m} + \dots + C_n^{n-m}$$

$$= C_{n-m+2}^{n-m+1} + C_{n-m+2}^{n-m} + \dots + C_n^{n-m}$$

$$= C_n^{n-m+1} + C_n^{n-m} = C_{n+1}^{n-m+1} = C_{n+1}^m = \binom{n+1}{m}$$

$$12) \binom{r+0}{0} \binom{m-0}{n-0} + \binom{r+1}{1} \binom{m-1}{n-1} + \dots + \binom{r+n}{n} \binom{m-n}{n-n}$$

$\binom{m+r+1}{n}$ 是从 $(0,0)$ 到 $(m-n+r+1, n)$ 的非降路径

按过 $x=r$ 上不同点 (r, k) 向右分类. $k=0, 1, \dots, n$

从 $(0,0)$ 到 (r, k) 非降路径有 $\binom{r+k}{k}$ 条

从 $(r+1, k)$ 到 $(m-n+r+1, n)$ 非降有 $\binom{m-k}{n-k}$ 条.

故从 $(0,0)$ 过 (r, k) 到 $(m-n+r+1, n)$ 有 $\binom{r+k}{k} \binom{m-k}{n-k}$

故由求和得 $\binom{m+r+1}{n}$

$$b) \sum_{k=0}^n C(2n, 2k)$$

$$① n=0. \quad \text{Bsp: } C_0^0 = 1$$

$$② n \neq 0 \quad \sum_{k=0}^n C_{2n}^{2k} = \frac{1}{2} \left(\sum_{k=0}^{2n} C_{2n}^k + \sum_{k=0}^{2n} (-1)^k C_{2n}^k \right)$$

$$\begin{aligned} \text{f) A) } (a+b)^n &= \sum C_n^k a^k b^{n-k} = \frac{1}{2} \left((1+1)^{2n} + (1-1)^{2n} \right) \\ &= \frac{1}{2} \cdot 2^{2n} = 2^{2n-1} \end{aligned}$$

$$\text{Teu. } \sum_{k=2}^{n-1} (n-k)^2 \binom{n-1}{k} = n(n-1) \cdot 2^{n-3} - (n-1)^2$$

$$\text{LHS} = \sum_{k=1}^n \underbrace{(n-k)^2}_{\substack{\text{geg.} \\ \downarrow}} \binom{n-1}{k} - (n-1)^2$$

$$= \sum_{k=1}^{n-1} \underbrace{k^2}_{\substack{\text{geg.} \\ \downarrow}} \binom{n-1}{k} - (n-1)^2$$

$$= (n-1)(n-2) \cdot 2^{n-3} - (n-1)^2$$

$$\sum_{k=0}^n k C_n^k = n \cdot 2^{n-1}$$

$$\sum_{k=0}^n k^2 C_n^k = n(n+1) 2^{n-2}$$

$$(n-k) C_{n-1}^{n-k} = (n-1) \underbrace{C_{n-1}^{n-k-1}}_{\substack{k \\ C_{n-1}^k}}$$

$$T_{24}(4) \quad \sum_{k=0}^n (-1)^k \frac{1}{k+1} \binom{n}{k} = \frac{1}{n+1}$$

$$k \binom{n}{k} = n \binom{n-1}{k-1}$$

$$\text{LHS} = \sum_{k=0}^n (-1)^k \frac{1}{k+1} \binom{n}{k}$$

$$= \sum_{k=0}^n (-1)^k \frac{1}{n+1} \binom{n+1}{k+1}$$

$$= \frac{1}{n+1} \sum_{k=0}^n \binom{n+1}{k+1} (-1)^k$$

$$\text{令 } k' = k+1$$

$$= -\frac{1}{n+1} \sum_{k'=1}^{n+1} \binom{n+1}{k'} (-1)^{k'}$$

$$= -\frac{1}{n+1} \left(\sum_{k'=0}^{n+1} \binom{n+1}{k'} (-1)^{k'} - 1 \right)$$

$$= -\frac{1}{n+1} \left((1-1)^{n+1} - 1 \right) = \frac{1}{n+1}$$

Pr2115

从 $S = \{\infty-0, \infty-1, \infty-2\}$ 中取 n 个数排列
不允许相邻位置数相同有几种排法

第1位数有3种. 第2位数有2种.

第3位则又有2种. 直至第 n 位.

故为 $3 \cdot 2^{n-1}$

T28 $S = \{n_1 \cdot a_1, n_2 \cdot a_2, \dots, n_k \cdot a_k\}$

求 S 各种 k -子集总和

选第 i 个 a_i 有 (n_i+1) 种

由乘法原理得 总数为 $N = \prod_{i=1}^k (n_i+1)$

Ex. $S = \{1 \cdot a_1, 1 \cdot a_2, \dots, 1 \cdot a_t, \infty a_{t+1}, \dots, \infty a_k\}$
 求 S 的 r 组合数

把 S 划分为 A 与 B 两集合

$$A = \{1 \cdot a_1, 1 \cdot a_2, \dots, 1 \cdot a_t\}$$

$$B = \{\infty a_{t+1}, \infty a_{t+2}, \dots, \infty a_k\}$$

从 A 中选 i 个. 从 B 中选 $r-i$ 个

$$\Downarrow$$

$$C_t^i$$

$$\Downarrow$$

$$C_{k-t+r-i-1}^{r-i}$$

$$\therefore N = \sum_{i=0}^r \binom{t}{i} \binom{k-t+r-i-1}{r-i}$$